

COURSE TITLE : SYNTHETIC RUBBERS
COURSE CODE : 3093
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75/3
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	General purpose synthetic rubbers.	21
2	Special purpose synthetic rubbers	21
3	Speciality synthetic rubbers	17
4	Thermoplastic elastomers and blends	16
TOTAL		75

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To understand the preparation , properties and applications of general purpose synthetic rubbers.
2	2	To understand the preparation , properties and applications of special purpose synthetic rubbers.
3	3	To know the preparation , properties and applications of speciality synthetic rubbers.
4	4	To comprehend the preparation , properties and applications of thermoplastic elastomers.
	5	To comprehend the preparation , properties and applications of blends.

SPECIFIC OUTCOME

MODULE I GENERAL PURPOSE SYNTHETIC RUBBERS

- 1.1.0. To Understand the preparation , properties and applications of general purpose synthetic rubbers.
- 1.1.1. Identify various general purpose synthetic rubbers.
- 1.1.2. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing , vulcanisate properties and applications of SBR.
- 1.1.3. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing , vulcanisate properties and applications of BR.
- 1.1.4 Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing , vulcanisate properties and applications of IR

- 1.1.5 Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of IIR.
- 1.1.6 Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of EPM and EPDM.
- 1.1.7 Mention the name and structure of diene in EPDM.

MODULE II SPECIAL PURPOSE SYNTHETIC RUBBERS

- 2.1.0. To understand the preparation, properties and applications of special purpose synthetic rubbers.
- 2.1.1. Identify various special purpose synthetic rubbers.
- 2.1.2. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of NBR.
- 2.1.3. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of CR.
- 2.1.4. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of CSM and Polysulphide rubbers.
- 2.1.5. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of Silicone rubber
- 2.1.6. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of polyurethane

MODULE III SPECIALITY SYNTHETIC RUBBERS

- 3.1.0. To know the preparation, properties and applications of speciality synthetic rubbers.
- 3.1.1. Identify various speciality synthetic rubbers.
- 3.1.2. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of fluoro carbon rubbers.
- 3.1.3. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of epichlorohydrin.
- 3.1.4. Explain the monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of acrylic rubbers.
- 3.1.5. Explain the monomer, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications of halobutyls.
- 3.1.6. Explain the monomer, polymerisation, properties and applications of carboxylated NBR, SBR and BR.
- 3.1.7. State the preparation, properties and application HTPB.
- 3.1.8. Explain the production, properties, and applications of hydrogels and ionomers.

MODULE IV THERMOPLASTIC ELASTOMERS AND BLENDS

- 4.1.0 To comprehend the preparation, properties and applications of thermoplastic elastomers.
- 4.1.1. Define thermoplastic elastomers..
- 4.1.2. State the advantages and disadvantages of TPE.
- 4.1.3. Describe the production, properties, compounding and applications of thermoplastic S-I-S, and S-B-S.
- 4.1.4. Describe the production, properties, compounding and applications of thermoplastic EVA

- 4.1.5. Describe the production, properties, compounding and applications of thermoplastic polyurethane.
- 4.1.6. Describe the production, properties, compounding and applications of thermoplastic polyester block co-polymer.
- 4.2.0 To comprehend the preparation , properties and applications of blends.
- 4.2.1. Define blending.
- 4.2.2. State the advantages of polymer blends.
- 4.2.3. Describe the dry blending and wet blending methods.
- 4.2.4. Explain the blending procedure, properties and application of NR/BR, NR/SBR, NR/BR/SBR.
- 4.2.5. Explain the blending procedure, properties and application of NR/EVA, NBR/PVC.

CONTENT DETAILS

MODULE I

General purpose synthetic rubbers

General purpose rubber-SBR, BR, IR, IIR, EPM and EPDM. Monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications- SBR, BR, IR, IIR, EPM, EPDM. Comparison –EPM, EPDM. Dienes- name, structure, properties.

MODULE II

Special purpose synthetic rubbers

Special purpose synthetic rubbers-NBR, CR, CSM, Silicone rubber, Polyurethane. Monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications- NBR, CR, CSM, Silicone rubber, Polyurethane.

MODULE III

Speciality synthetic rubbers

Speciality synthetic rubbers-Fluorocarbon, epichlorohydrin, acrylic , halobutyls. Monomer, monomer preparation, polymerisation, raw polymer properties, compounding, curing, vulcanisate properties and applications- Fluorocarbon, epichlorohydrin, acrylic , halobutyls. Carboxylated NBR, SBR, BR- preparation – properties and application. HTPB- preparation – properties and application. Hydrogels- ionomers, preparation, properties and application.

MODULE IV

Thermoplastic elastomers and blends

Definition- advantages and disadvantages of TPE. Production properties, compounding, application of S-I-S, S-B-S, EVA, PU and Polyester block copolymer. Polymer blends- definition, advantages. Blending procedure, properties application- NR/BR, NR/SBR, NR/BR/SBR, NR/EVA, NBR/PVC.

REFERENCE BOOKS

1. Synthetic rubbers-D.C.Blackley (Publisher – Springer Netherlands Edition 1)
2. Rubber Technology - Mauris Morton(Publisher – Springer Netherlands 1999 Edition)
3. Vanderbilt Rubber hand book - R.T Vanderbilt (Publisher- R.T Vanderbilt 1990 Edition)